PATHWAYS BETWEEN BOROUGHS: A NETWORK MODEL OF THE NEW YORK CITY SUBWAY SYSTEM

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Abstract. The New York City subway system is widely regarded as one of the most extensive transit networks in the world, consisting of 472 stations, 24/7 service, and coverage across all five of the city's boroughs. Its iconic map symbolizes the city's connectivity, but does not fully reflect the variation in accessibility and connectivity across boroughs. Variations in overcrowding, service delays, and geographic accessibility contribute to differing levels of mobility throughout the city. This premise raises an important question: How can such a large and interconnected system produce such disparate transit experiences for riders across communities?

This paper addresses this question by constructing a spatial network of New York City community districts based on shared subway line connectivity. The network models NYC community districts as nodes and establishes a shared edge when a subway line serves both districts. To analyze and evaluate relative connectivity and accessibility within and between districts, various centrality measures, such as weighted degree and betweenness centrality, are computed. These measures identify districts that are highly connected and are strategically important to the network. The results show that connectivity is concentrated in Manhattan and parts of Brooklyn, with these areas serving as major hubs. While Queens and the Bronx have more dispersed coverage. Surprisingly, Staten Island stands as the most disconnected borough. These findings reveal that the infrastructure of the NYC subway system shapes the uneven patterns of mobility and accessibility experienced by riders across districts. More broadly, these results offer insights relevant to urban planning, transit equity, and economic geography.

1. Introduction

The New York City subway system has attracted increased attention due to concerns about riders' experience, including overcrowding, frequent train delays, and unequal access. Urban transportation systems play a critical role in shaping connectivity and accessibility within cities. Subway networks, in particular, serve as key infrastructure for linking neighborhoods, supporting mobility, and facilitating economic and social activity. Understanding how transit systems link sectors of a city or across a variety of neighborhoods can provide insights into spatial inequalities, accessibility patterns, and the broader structure of urban environments.

Transportation networks are commonly modeled as networks in which stations or transit stops serve as nodes, with routes connecting them as edges. In our network representation of New York City's subway system, nodes correspond to community districts and edges represent shared subway lines between districts. Specifically, an edge is created between two districts when at least one subway line serves stations in both districts. Edge weights reflect the number of subway lines shared between districts, thereby distinguishing stronger transportation relationships from weaker ones. Community districts provide a useful unit of analysis because they capture meaningful local variation while still allowing meta-comparison across the five boroughs. This enables evaluation of connectivity not only at the citywide scale but also across boroughs and between districts with varying levels of activity and accessibility. Using network-based measures of centrality and connectivity, this paper identifies districts of great significance within the network, districts (or communities) that act as transportation hubs, and lines that span the largest amount of New York City. We also consider the impact of established transit hubs and prominent NYC destinations as an explanation for districts that emerge as more central than others.

2. Background

Urban transportation systems play a central role in shaping how cities function by influencing access to employment, education, public services, and recreation. In large metropolitan areas, transit infrastructure does more than facilitate population movement; it also helps shape patterns of economic activity, neighborhood development, and spatial accessibility. Because access to reliable transportation is uneven, the study of transit systems can reveal broader inequalities in how communities are connected within an urban environment. For this reason, transportation networks are often analyzed not only as physical systems, but also as social and spatial frameworks that shape everyday urban life.

New York City provides an especially important setting for this type of analysis because its subway system is among the largest and most complex transportation networks in the world. The system is housed in the home of 8.48 million people, servicing millions of them every day, and links residential areas to major commercial, cultural, and recreational destinations. To identify patterns in connectivity and accessibility, this project uses community districts as the main unit of analysis. Community districts offer a useful middle scale as they are broader than individual stations but more detailed than borough-level analysis alone. This allows for comparison between local areas while still capturing citywide structure. Using network analysis, districts can be represented as nodes and shared subway line intersections as edges, enabling the evaluation of which districts are most connected, which serve as bridges between others, and which occupy the most central positions in the overall system. Measures such as weighted degree, degree centrality, and betweenness centrality quantify different forms of importance.

Degree Centrality

$$x_i = k_i \tag{1}$$

where x_i represents the degree centrality of node i, and k_i denotes the number of edges connected to that node. In our model, a district with a high degree centrality shares subway line connections with many other districts, indicating a high level of direct accessibility.

Weighted Degree Centrality

$$s_i = \sum_{j=1}^{n} w_{ij} \tag{2}$$

where s_i represents the weighted degree of node i, and w_{ij} denotes the weight of the edge between nodes i and j. Weighted degree extends degree centrality by incorporating edge weights into the calculation. In this network, edge weights correspond to the number of subway lines shared between specific districts. Districts with high weighted degree therefore maintain not only many connections, but also stronger transportation relationships across our network.

Betweenness Centrality

$$b_i = \sum_{s=1}^{n} \sum_{t=1}^{n} \frac{n_{st}^i}{g_{st}} \tag{3}$$

where b_i represents the betweeness centrality, g_{st} is the number of shortest paths between nodes s and t, and n_{st}^i is the number of those paths that pass through node i. Betweenness centrality measures how frequently a node lies on the shortest paths between other nodes in the network. Districts with high betweenness centrality function as important bridges that facilitate movement between different districts of the city.

Despite its scale and coverage, subway access is not evenly distributed across NYC. Manhattan contains a dense concentration of stations and transfer points, whereas other boroughs, such as Queens, the Bronx, and Staten Island, have more limited or spatially dispersed service. This unevenness makes NYC a strong setting for further examining why transit connectivity varies across space and why some districts become more central than others within the larger urban network. We consider the role of major destinations and points of interest in shaping connectivity patterns. Well-known sites such as Charging Bull, Brooklyn Flea, Luna Park, and Grand Central Terminal illustrate how cultural, commercial, and recreational landmarks are distributed across the city and may help explain why some districts attract greater movement and appear more central in the network. Examining transit connectivity alongside these destinations links the structural properties of the subway system to the lived geography of the city, revealing how transportation and urban activity reinforce one another.

3. Our Model

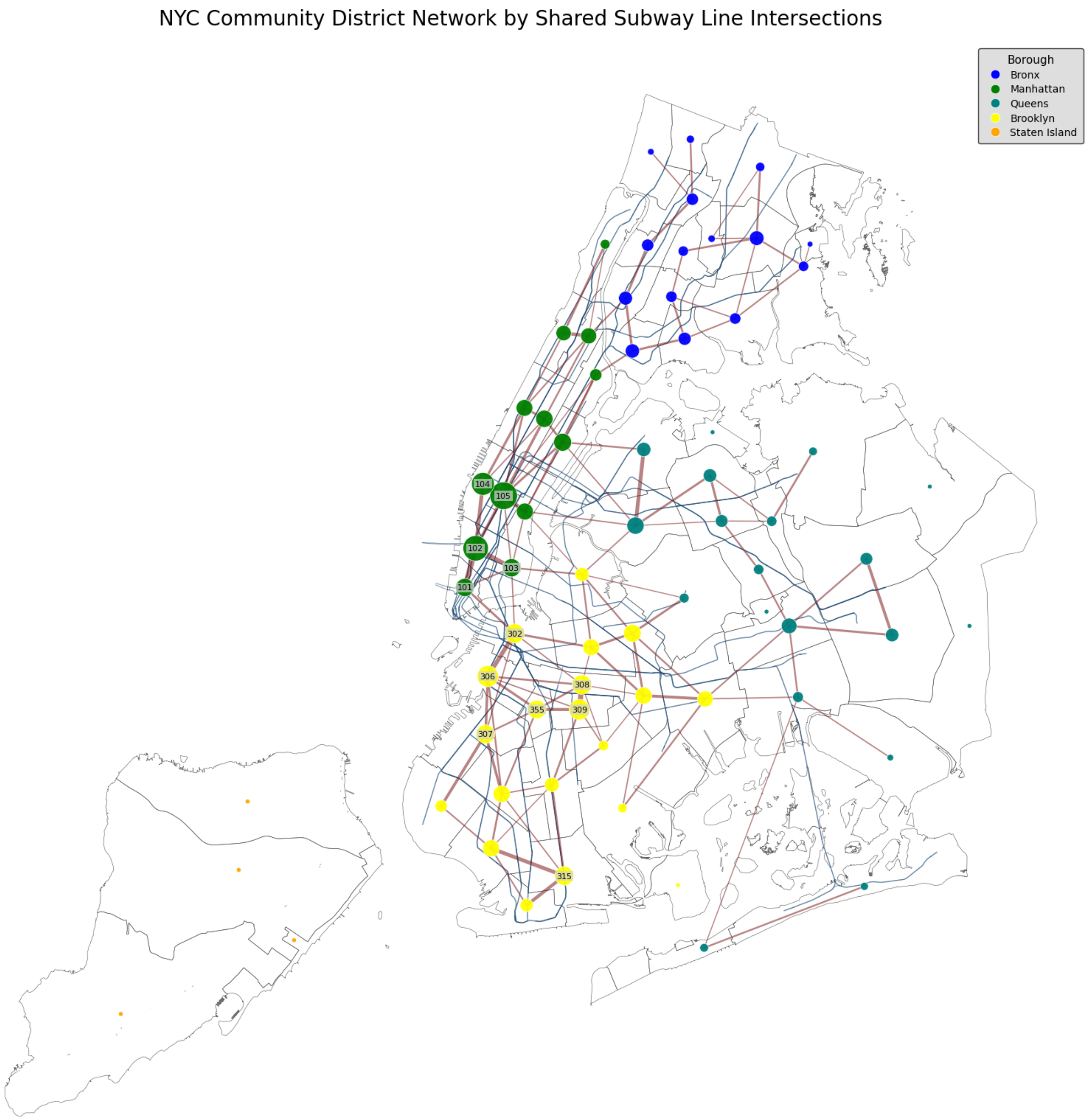

Figure 1. NYC Subway Connectivity

The construction of our network and supplemental materials was completed in Google Colaboratory (Colab), where we used multiple Python scripts to clean geographic data, create maps, build network graphs, and calculate centrality measures. The major datasets used in this project included NYC borough boundary data, NYC subway station data, NYC subway line data, NYC community district boundary data, subway line intersections with community districts, and district-level demographic information collected from the U.S. Census Bureau.

Geographic data were gathered primarily through Overpass Turbo and OpenStreetMap, which required learning how to write queries that were both accurate and simple enough to return only the relevant features, such as MTA subway stations, subway lines, borough boundaries, and community districts, while excluding unnecessary records such as bus routes, service tracks, yards, sidings, and the Staten Island Railway. Borough and community district polygons were also referenced through boundaries.beta.nyc and read into GeoPandas as GeoJSON layers to create the base NYC map used throughout the analysis. In the coding workflow, borough files were standardized so that the five boroughs were consistently labeled as Bronx, Brooklyn, Manhattan, Queens, and Staten Island; duplicate subway station points were removed by station name to improve the reliability of station counts; and subway line files were cleaned so that only the main NYC subway network was included.

Community districts were treated as nodes, and edges were created when the same subway line intersected more than one district; if districts shared multiple subway line intersections, the connecting edge received a higher weight. This allowed us to move from a simple map of subway coverage to a network in which node size represented the strength of subway-linked connectivity, node color represented borough, and edge weight reflected repeated shared subway line intersections.

We then calculated three main measures of network importance:

(1) Weighted Degree
(2) Degree Centrality
(3) Betweenness Centrality

Using weighted degree as the primary measure because it best captures how strongly a district is connected within the subway system. Since no members of our team had a strong programming background at the start of the project, a major part of the process involved relying on class resources to develop foundational coding strategies and using SuarezAI to help polish the Python code, debug errors, and generate more comprehensive graphics. Even with that support, the project required substantial independent work to locate usable data, learn query-writing in Overpass Turbo, and transform raw census and geographic information into a working network model of subway connectivity across New York City.

4. Results

4.1. **Network Overview.** Manhattan contains 12 community districts situated very close to one another. Its densely clustered districts are served by subway lines that link many other districts. These network features prop Manhattan to appear as the central borough of the network. Brooklyn contains 18 districts that spatially span more of the city than those of Manhattan, while being relatively close to one another across the borough. Brooklyn districts are connected by subway lines that link many other districts, as indicated by the large size of select nodes, especially in areas closer to Manhattan and downtown Brooklyn. Queens contains 14 districts, which span the largest portion of New York City. In turn, its districts are situated far from one another. Districts closer to Manhattan and downtown Brooklyn have subway lines that connect to many other districts, whereas the borough's more peripheral districts appear smaller, with some containing notably 2 districts that lack subway lines connecting to neighborhoods beyond their own boundaries. Bronx County comprises 12 districts that span a substantial portion of the city. Its districts, like Brooklyn's, are closely situated. Districts closer to Manhattan have subway lines that connect to many other districts, while its peripheral districts don't. Unlike Queens, all of the Bronx's districts are connected to at least one other district, enabling interconnected mobility within and beyond the district. Finally, Staten Island comprises simply 3 districts. Neither of the borough's districts is connected to the other, and the borough as a whole is completely disconnected from the other boroughs. Staten Island appears completely disconnected because it is not served by the main NYC subway system (The MTA). This visualization supports one of our main findings that subway connectivity is not evenly distributed across NYC.

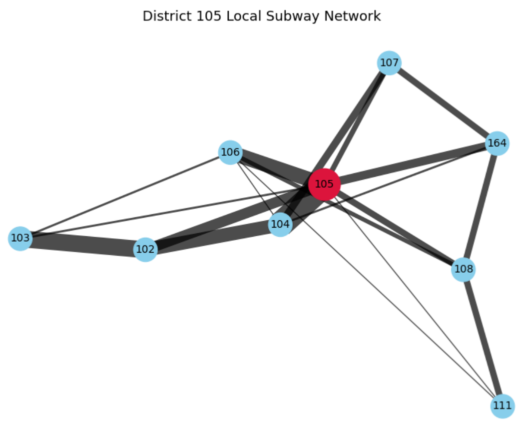

Figure 2. Times Square Connectivity

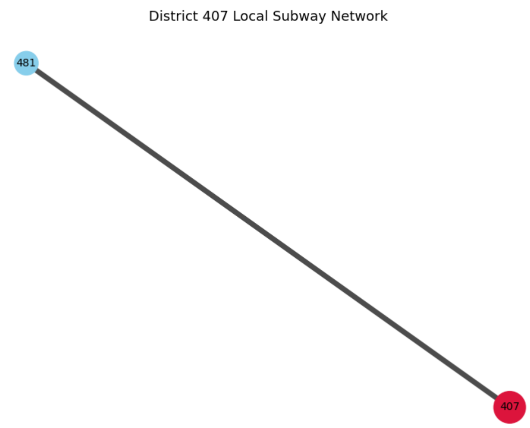

Figure 3. College Point Connectivity

District 105, located in Manhattan, is the most connected district in the city. This district is particularly notable because it houses Times Square, a major hub of urban activity, and, in turn, subway service is reinforced by nearby tourism infrastructure, such as hotels. In contrast, District 407, located roughly in the College Point area of Queens, is far more isolated, connecting to a single adjacent district rather than functioning as a broader hub. Overall, this graphic shows that subway-based accessibility is concentrated in a limited set of core districts rather than being evenly distributed across the city. It's important to note that this pattern reflects only the MTA subway system; it does not extend to other forms of public transportation that also serve New York City, as our study's focus does not include them.

Top 15 Most Subway-Linked Community Districts
(Ranked by Weighted Degree)

District ID	District	Borough	Weighted Degree	Degree Centrality	Betweenness
105	105	Manhattan	70	0.114	0.095
102	102	Manhattan	59	0.057	0.0
104	104	Manhattan	47	0.071	0.127
306	306	Brooklyn	41	0.086	0.001
309	309	Brooklyn	38	0.071	0.013
308	308	Brooklyn	34	0.1	0.198
302	302	Brooklyn	33	0.086	0.132
315	315	Brooklyn	32	0.057	0.0
307	307	Brooklyn	31	0.071	0.018
355	355	Brooklyn	30	0.071	0.0
103	103	Manhattan	29	0.086	0.15
101	101	Manhattan	29	0.043	0.0
108	108	Manhattan	28	0.086	0.025
304	304	Brooklyn	27	0.071	0.066
402	402	Queens	26	0.086	0.163

Figure 4. Top Connected Districts Centrality Measures

4.2. **Centrality Overview.** The table displays the top 15 NYC community districts ranked by weighted degree. As our study aims to draw conclusions about connectivity, this measure aligns best with our project goals. Weighted degree is the most important metric in this report because it shows how strongly a district is connected via subway lines. A district with a higher weighted degree is not just connected to more geographical places but may also share multiple subway line connections with other districts. Districts with high weighted degrees help move people across the city. If these districts experience delays, crowding, or closures, the effects can spread across the entire network. In this way, highly weighted degree districts are both strengths and have major drawbacks. The system is efficient and connected, but disruptions in these areas can affect many other routes daily.

The top 5 districts by this measure are 105, 102, 104, 306, and 309.The top 3 districts are located in Manhattan while the remaining are in Brooklyn. District 105 has the highest weighted degree at 70, making it the most linked district in the network. District 102 is second with 59, and District 104 is third with 47. This shows a high connectivity of Manhattan districts to the larger subway system.Brooklyn districts' appearance in the top 5 rankings suggests that Brooklyn is not just an outer borough connected to Manhattan, it has its own level of connectivity within the network. A Queens network appears once in this list of top districts, with District 402 having a weighted degree of 26. This suggests that Queens still has an important linked districts, but fewer than in Manhattan and Brooklyn.

Degree centrality measures how many direct connections a node has compared to all other nodes in the network. District 308 in Brooklyn has a degree centrality of 0.100, one of the highest within out table. This suggests that this Brooklyn district serves as a direct link to other districts. This probes the existence of districts as a hub. While Manhattan has a high weighted degree, Brooklyn has a stronger direct connectivity. This illustrates that some Brooklyn districts also play major roles in connecting routes and neighborhoods in NYC. Degree centrality helps explain how accessible a district is within our network. A district with many direct connections offers riders more route options and fewer transfers.

Betweenness centrality measures whether a node acts as a bridge or connection between other nodes. A district with high betweenness centrality may not have the most connections, but are an important connection between different parts of the network. District 308 in Brooklyn has a betweenness value of 0.198, the highest among the top 15 while District 302 in Brooklyn has a betweenness of 0.132. This show that some districts act as bridges between different districts of the city. Meaning that transit access and importance are not only about having the most stations or the most subway lines. Some districts matter because they connect otherwise separate parts of the network. These are the areas where transfers and traveling through boroughs become important.

5. Conclusions and Discussion

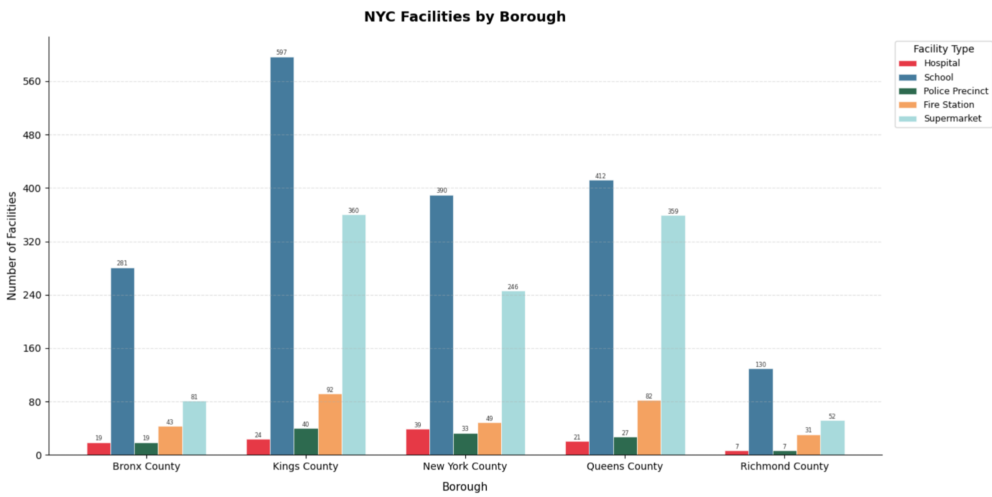

Figure 5. Facilities by Borough

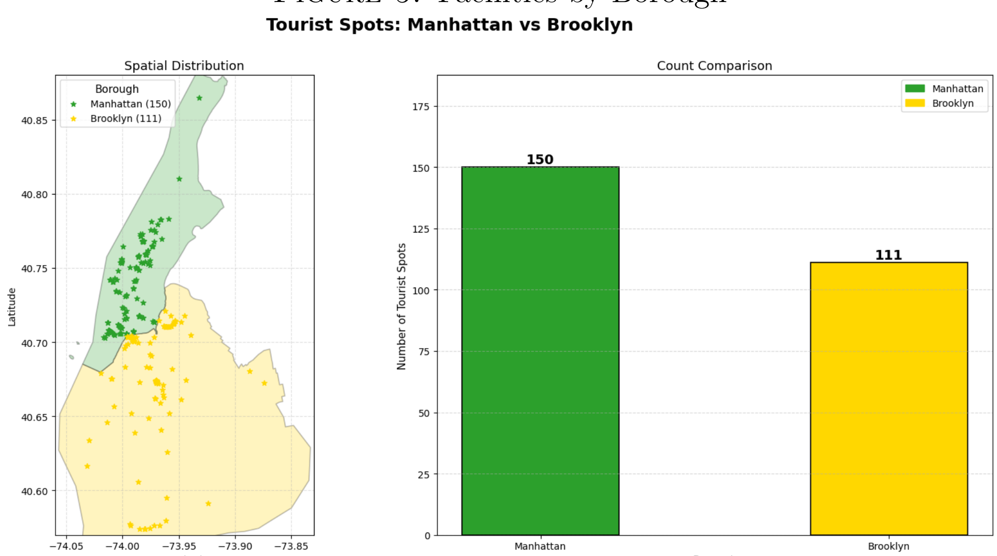

Figure 6. Tourist Spots: Manhattan vs BK

Our results suggest that network centrality in New York City is shaped not only by the number of everyday facilities in a borough, but also by the concentration of major destinations that attract movement across the city. Although the facilities charts show that

Brooklyn (Kings County) contains more total facilities than Manhattan (New York County), especially in categories such as schools, fire stations, and supermarkets, the tourism chart indicates that Manhattan has a greater concentration of tourist and destination-oriented sites overall, with 150 identified tourist spots compared to Brooklyn's 111. This helps explain why Manhattan appears more central in the subway network: it contains a denser cluster of highly visited destinations that generate repeated trips, transfers, and cross-borough travel. In addition to its role as the historic and commercial core of the city, Manhattan includes prominent destinations such as SoHo, Times Square, and Union Square, all of which reinforce its importance as a hub of movement. Brooklyn also contains major attractions, including Coney Island and well-known neighborhood destinations like Dumbo, but these appear to be more spatially dispersed and less concentrated. Taken together, the charts suggest that Manhattan's higher network connectivity is not simply a matter of having more infrastructure, but rather of having a stronger concentration of destinations that pull riders into the borough and through it, making Manhattan a more central and connected core within the subway system, therefore making it practical to have more subway transportation within the borough.

5.1. **Limitations and Future Improvements.** This study has several important limitations. First, the analysis is based on geographic intersections between subway lines and community districts, rather than actual rider behavior. As a result, a district may appear highly connected in the network while experiencing relatively low ridership in practice due to factors such as time of day, service frequency, or station conditions. Second, the network is constructed at the community district level rather than the station level. Consequently, major transit hubs such as 14th Street, Times Square, and Grand Central Terminal are not individually represented or ranked in the centrality analysis. Instead, their influence is only indirectly captured through the districts in which they are located. Third, data cleaning decisions influence the structure of the network. In particular, the exclusion of service tracks and the Staten Island Railway simplifies the analysis but also contributes to the apparent isolation of Staten Island, which is not integrated into the primary subway system.

The network could be reconstructed at the station level, with individual subway stations as nodes and connections or transfers as edges. This would allow for direct centrality rankings of major hubs such as 14th Street–Union Square, Grand Central Terminal, Times Square–42nd Street station, Fulton Street station, and Atlantic Avenue–Barclays Center station. Second, incorporating ridership data would allow for a direct comparison between network centrality and passenger volume. This would help determine whether the most structurally central locations in the network are also the busiest in practice.

Future work could identify potential areas for transit expansion by combining measures of population density, network centrality, and station accessibility. Districts characterized by high population, low connectivity, and limited proximity to subway stations could be prioritized as areas of need.

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Appendix A. Group Contributions

Esther Osibajo: Editing, Writing, Interpretation of Results
Daniel Lucero: Coding, Data Cleaning and Visualization, Network Building
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